system.' It is also called as grid. In this system, the power plant and sub-station are connected together, and the voltage is reduced to 33KV, by using transformer. This method increases the reliability and efficiency. Electricity can be provided from different power plants during high power consumption.

1.10.4 Service lines

Service line is low length connecting conductor. These service lines act as conductor between distribution pole and consumer.

1.11 Underground cables

By using underground cable, the power is transmitted from the generating station to the consumers. When electrical power is unable to transmit by overhead lines or in a thickly populated area, underground cables are used. An underground cable consists of one or more conductor covered with suitable insulation and surrounded by a protecting core. Normally the number of cores in underground cables are 1, 2, 3, 3 ½ and 4.

1.11.1 Advantages of cables

i. Underground cables are not subjected to lightning discharges, thunder, storms, birds and other severe weather conditions.
ii. As there is no tension on the conductor, it will not break due to mechanical reasons after being installed. Hence there are very few chances for a power failure. It reduces accidents.
iii. Less maintenance is required.
iv. There is no interference with tele communication circuits.

1.11.2 Disadvantages of cables

i. The installation cost is high.
ii. High voltage cables are difficult to manufacture due to the insulation problem.
iii. Joining of underground cables is difficult.
iv. Fault location is not easy.

1.11.3 Classification of cables

a. According to the rating of voltage the cables are classified into
   i. Low tension cables up to 1 KV
   ii. High tension cable up to 11KV
   iii. Super tension cables 22 KV to 33KV
   iv. Extra high-tension cables 33KV to 66 KV
   v. Oil filled cables 66KV to 132 KV
   vi. Extra super voltage cables beyond 132 KV
b. According to insulation, cables are classified into
   i. PVC insulated cables
   ii. Mineral insulated cable
   iii. Paper insulated lead sheathed cable
   iv. Cross linked poly ethylene cable
   v. Paper insulated lead covered double tap armored cable
c. According to number of conductors, cables are classified into
   i. Single core cable
   ii. 2 core cable
   iii. 3 core cable

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